What is MTOM?

MTOM (Message Transmission Optimization Mechanism) allows web services to send binary conte nt, such as images or files, without converting it to base64 text within the SOAP envelope. Normally, binary data is converted to xs:base64Binary in XML, which increases the data size by approximately 3 3% and adds processing overhead. MTOM overcomes this by transmitting binary data as separate MI ME attachments while keeping it logically part of the SOAP message.

Web service Code:

```
/*
 * To change this template, choose Tools | Templates
 * and open the template in the editor.
 */
package tycs;
import java.io.*;
import javax.jws.Oneway;
import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;
import javax.xml.ws.soap.MTOM;
/**
 *
 * @author Administrator
 */
@MTOM(enabled=true, threshold=6000)
@WebService(serviceName = "imagews1")
public class imagews1 {

    /**
     * This is a sample web service operation
     */
    @WebMethod(operationName = "hello")
    public String hello(@WebParam(name = "name") String txt) {
        return "Hello " + txt + " !";
    }

    /**
     * Web service operation
     */
    @WebMethod(operationName = "upload")
    @Oneway
    public void upload(@WebParam(name = "Filename") String Filename, @WebParam(name =
"ImageBytes") byte[] ImageBytes) {
        String filePath = "C:\\Users\\Administrator\\Downloads\\" +Filename;
        try{FileOutputStream fos = new FileOutputStream(filePath);
              BufferedOutputStream bos = new BufferedOutputStream(fos);
              bos.write(ImageBytes);
     System.out.println("Received file:" +filePath);
    } catch (Exception ex) {
    System.out.println(ex);
```

```
}
}

    /**
     * Web service operation
     */
    @WebMethod(operationName = "download")
    public byte[] download(@WebParam(name = "Filename") String Filename) {
        String filePath = "C:\\Users\\Administrator\\Downloads\\" +Filename;
        System.out.println("Sending file:" +filePath);
        try{FileInputStream fis = new FileInputStream(new File(filePath));
        BufferedInputStream bis = new BufferedInputStream(fis);
            byte[] fileBytes = new byte[(int) new File(filePath).length()];
            bis.read(fileBytes);
            return fileBytes;
        } catch (Exception ex) {
            System.out.println(ex);
        }
        return null;
        }

    }
```

INDEX.JSP CODE:

```
<%--

    Document   : index

    Created on : 3 Aug, 2026, 11:41:47 AM

    Author     : Administrator

--%>


<%@page contentType="text/html" pageEncoding="UTF-8"%>

<!DOCTYPE html>

<html>

    <head>

        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">

        <title>JSP Page</title>

    </head>

    <body>

        <h1>Hello World!</h1>
```

```
    <%
try {
    cs.Imagews1_Service service = new cs.Imagews1_Service();
    cs.Imagews1 port = service.getImagews1Port();

    javax.xml.ws.BindingProvider bp = (javax.xml.ws.BindingProvider) port;
    javax.xml.ws.soap.SOAPBinding binding = (javax.xml.ws.soap.SOAPBinding) bp.getBinding();
    binding.setMTOMEnabled(true);

    String action = request.getParameter("action");
    String filename = request.getParameter("filename");
    if ("upload".equals(action) && filename != null) {
        java.io.File file = new java.io.File("C:\\Users\\Administrator\\Downloads\\" + filename);
        if (file.exists()) {
            byte[] bytes = new byte[(int) file.length()];
            java.io.FileInputStream fis = null;
            try {
                fis = new java.io.FileInputStream(file);
                fis.read(bytes);
            } finally {
                if (fis != null) { fis.close(); }
            }
            port.upload(filename, bytes);
            out.println("<p style='color:green;'>Uploaded successfully via MTOM!</p>");
        } else {
            out.println("<p style='color:red;'>File not found in Downloads folder!</p>");
        }
    }

    if ("download".equals(action) && filename != null) {
        byte[] imgBytes = port.download(filename);
```

```
            if (imgBytes != null && imgBytes.length > 0) {
                String base64Str = javax.xml.bind.DatatypeConverter.printBase64Binary(imgBytes);
                out.println("<h3>Downloaded Image:</h3>");
                out.println("<img src='" + base64Str + "' width='300'/>");
            } else {
                out.println("<p style='color:red;'>File not found on server.</p>");
            }
          }
        } catch (Exception e) {
            out.println("<p style='color:red;'>Error: " + e.getMessage() + "</p>");
        }

    %>
        <hr/>
    <form action="index.jsp" method="POST">
        Enter Filename:
        <input type="text" name="filename" placeholder="test.jpg" required>
        <br/><br/>
        <button type="submit" name="action" value="upload">Upload File</button>
        <button type="submit" name="action" value="download">Download File</button>
    </form>

    </body>
</html>
```

OUTPUT:

NOTE: While downloading and uploading the file name should be same.

Hello World!

Uploaded successfully via MTOM!

Enter Filename: test.jpg

Upload File | Download File

Hello World!

Downloaded Image:



Enter Filename: test.jpg

Upload File | Download File